

HFSCIM: Hybrid FeFET-SRAM CIM Architecture for Continuous-Time Inference and Online Reinforcement Learning

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Abstract

Edge reinforcement learning (RL) is essential for continuous robotic control yet suffers from high latency and energy consumption. Compute-in-memory (CIM) emerges as a promising hardware solution for real-time online learning. However, existing solutions suffer from three fundamental limitations: (1) discrete-time RL introduces truncation error and mismatches real-world continuous dynamics; (2) conventional CIM architectures (e.g., FeFET, SRAM) trade density, precision, and efficiency, making them unsuitable for high-fidelity control; and (3) enabling online training is difficult due to non-volatile memory endurance constraints and extra storage overhead for volatile memory. We present *HFSCIM* (Hybrid FeFET-SRAM CIM), an architecture co-designed with a continuous-time Soft Actor-Critic (CT-SAC) algorithm based on Neural Ordinary Differential Equations. CT-SAC eliminates discretization artifacts, enabling smooth and physically consistent control trajectories. The *HFSCIM* macro combines energy-efficient FeFET analog computation with adjustable-precision SRAM-based Low-Rank Adaptation (LoRA) noise compensation, achieving both high precision and compact area. To support online adaptation, we introduce a FeFET-Friendly online learning scheme that performs frequent LoRA-based updates in SRAM and infrequent FeFET programming, overcoming device-level variability and endurance limitations. Extensive experiments demonstrate that *HFSCIM* minimizes the frequency of FeFET weight updates by 95.0%, while delivering 25.4× to 310.3× speedups and 3.3× to 43.5× energy efficiency improvements over state-of-the-art hardware baselines. *HFSCIM* offers a complete hardware-software co-design blueprint for continuous-time, high-efficiency RL on resource-constrained systems.

CCS Concepts

• **Hardware** → **Memory and dense storage.**

Keywords

Compute-in-memory, reinforcement learning, neural ODE, homogeneous architecture.



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1 Introduction

Edge robots increasingly rely on reinforcement learning (RL) to adapt their behavior through continual interaction with the physical world [25, 27]. However, deploying RL on resource-constrained platforms remains fundamentally challenging because the algorithmic assumptions [23], memory technologies [31], and compute architectures [6] used in conventional systems are mismatched with the requirements of real-time continuous robotic control. These mismatches translate into degraded policy performance, excessive power consumption, and limited online learning capability. Compute-in-memory (CIM) provides a promising solution for edge reinforcement learning [14, 21, 22, 28, 33, 35, 36]. However, deploying both inference and learning on resource-constrained edge platforms still faces the following three challenges (Figure 1 Upper):

Challenge 1: Discrete-time RL poorly matches continuous physical dynamics. Most modern RL algorithms adopt discrete-time policies and assume a fixed control time step, despite the fact that real-world dynamics evolve continuously. These formulations are inherently sensitive to the chosen time step [23] and introduce truncation errors that accumulate over long trajectories. Recent efforts such as [18] attempt to mitigate discretization sensitivity, but they remain constrained by their discrete-time nature and still suffer information loss. This algorithm-dynamics mismatch fundamentally limits control fidelity and hinders stable policy optimization.

Challenge 2: Homogeneous CIM faces a precision-area inference dilemma. Ferroelectric field-effect transistor (FeFET)-based CIM crossbars [1, 11, 16, 19, 20] provide high density and analog parallelism but suffer from limited precision and programming noise, which restrict their reliability for large-scale neural policies. Conversely, Static Random Access Memory (SRAM)-based CIM [2, 7, 29] achieves high precision and better computational linearity, yet suffers from significantly larger cell area and reduced storage density. Neither pure FeFET nor pure SRAM CIM provides the right balance of precision, energy efficiency, and area efficiency required for continuous-time control running entirely at the edge.

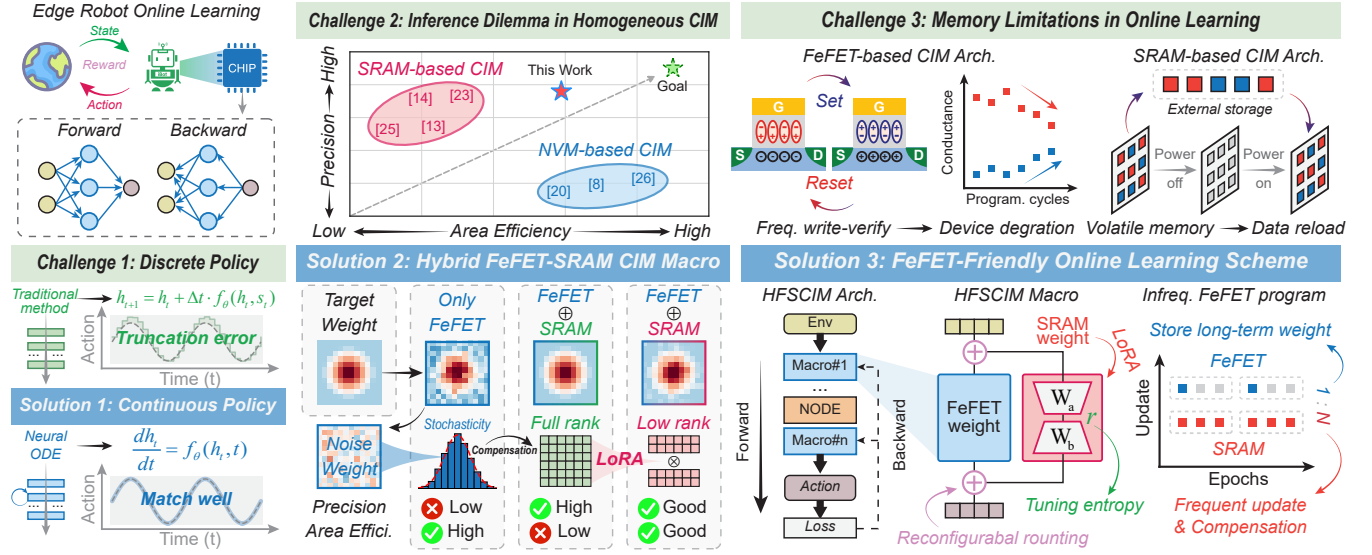


Figure 1: Three main challenges and the proposed HFSCIM architecture for continuous-time policy addressing low algorithm capability, inference precision, and online learning efficiency for edge reinforcement learning.

Challenge 3: Enabling efficient online learning is challenging for both FeFET CIM and SRAM CIM. Reinforcement learning on the edge requires on-device weight updates, but implementing training inside CIM arrays is especially challenging. FeFET memories suffer from limited write endurance and high programming energy, making frequent stochastic gradient descent (SGD) updates impractical [34]. SRAM-based CIM supports fast and accurate updates but is volatile and area-expensive, forcing costly off-chip back-up storage [4]. These device-level limitations directly constrain the feasibility of edge RL, where thousands to millions of gradient updates are required.

Our Solutions. To overcome these three challenges, we propose *HFSCIM*, a co-designed software-hardware framework that maps continuous-time RL policy onto a hybrid FeFET–SRAM CIM architecture coupled with a FeFET-Friendly online learning scheme (Figure 1 Bottom). The innovations are as follows:

- **Solution 1: Continuous-time policy co-design via neural ordinary differential equation (ODE) solver.** We propose the first software-hardware co-design integrating a fully analog neural ODE solver with a continuous-time Soft Actor-Critic (CT-SAC). Software-wise, we introduce a Neural ODE formulation of CT-SAC that models policy evolution by learning the underlying vector field rather than discrete updates, generating physically consistent trajectories aligned with real-world dynamics. Hardware-wise, these policy outputs drive analog integrators to realize neural ODE dynamics via physical RC evolution, eliminating discrete numerical steps.
- **Solution 2: Hybrid FeFET–SRAM CIM macro.** We design a hybrid-memory CIM macro featuring dense FeFET arrays for ultra-efficient analog vector-matrix multiplication (VMM). We further integrate an SRAM branch based on compact Low-Rank Adaptation (LoRA) [10] to provide flexible configurations for adjustable-precision noise compensation,

effectively balancing the density-precision trade-off inherent to homogeneous memories. The HFSCIM macro also supports both feed-forward (FF) and backpropagation (BP) via reconfigurable routing, underpinned by highly linear analog adders that facilitate hybrid charge- and current-domain computations.

- **Solution 3: FeFET-Friendly online learning.** We introduce a FeFET-Friendly online learning scheme where frequent low-rank updates and compensation occur exclusively in SRAM, while infrequent one-shot programming to the FeFET array preserves long-term weights, thereby circumventing endurance limitations and ensuring stable training under analog variability. The tunable SRAM rank enables dynamic precision scaling to satisfy RL exploration entropy requirements.
- **Comprehensive validation.** We evaluate the proposed CT-SAC and HFSCIM co-design across four RL environments, demonstrating improvements in both algorithmic performance and hardware efficiency. Software-wise, CT-SAC achieves a 9.70% to 20.60% improvement in cumulative rewards over state-of-the-art baselines. Hardware-wise, the HFSCIM architecture delivers up to a 310.3× speedup and a 43.5× improvement in energy efficiency over digital counterparts. Under device variability, the macro yields an 18.7 dB signal-to-noise ratio (SNR) for vector-matrix multiplication, nearly doubling the 9.3 dB of standard FeFET crossbars. Compared to conventional SRAM and FeFET CIM, our approach reduces write overhead, cutting average parameter updates by 90.9% and specific FeFET weight updates by 95.0%.

2 Preliminary

2.1 Online Reinforcement Learning

Standard online RL learns a policy π that maximizes the expected cumulative return J_π from interactions stored in a replay buffer D :

$$J_\pi = \mathbb{E}_\pi \left[\sum_{i=t}^{\infty} \gamma^{i-t} r(s_i, a_i) \right], \quad (1)$$

where γ is the discount factor and $r(s_i, a_i)$ is the reward.

For continuous-control tasks, model-free algorithms such as Deep Deterministic Policy Gradient (DDPG), Proximal Policy Optimization (PPO), Twin Delayed DDPG (TD3), and Soft Actor-Critic (SAC) are widely used [9, 26]. SAC is particularly relevant to this work because it jointly optimizes a critic for value estimation and an entropy-regularized actor for exploration, with the temperature α tuned toward a target entropy $\mathcal{H}$.

2.2 Neural Ordinary Differential Equation

Neural ODEs model hidden-state evolution by replacing discrete network layers with continuous dynamics [3, 5, 8, 32]. A neural network f_θ defines the time derivative of a latent state $h(t)$:

$$\frac{dh(t)}{dt} = f_\theta(h(t), t) \quad (2)$$

The state at $t + dt$ is obtained by integrating this vector field:

$$\text{ODESolve}(f_\theta, h(t), dt) = h(t) + \int_t^{t+dt} f_\theta(h(t), t) dt. \quad (3)$$

2.3 FeFET/SRAM-based CIM Basics

FeFET-based CIM. FeFETs store weights through ferroelectric polarization states in the gate, enabling high-density non-volatile CIM (nvCIM) operations with low leakage and parallel analog computation [15]. However, analog polarization switching introduces conductance variation, while high-voltage programming increases write energy and constrains endurance [12].

SRAM-based CIM. SRAM-based CIM uses 6T or 8T bitcells for in-memory digital computation [21, 33]. Cross-coupled inverters and CMOS access transistors provide deterministic switching and high precision, but peripheral circuits and large bitcells incur substantial area overhead and lower density than nvCIMs.

3 Software-Hardware Co-Design

To achieve high-fidelity and energy-efficient continuous control on resource-constrained edge hardware, we propose *HFSCIM*, a software-hardware co-design architecture tailored for continuous-time inference and online learning. The proposed framework encompasses the CT-SAC algorithm based on a neural ODE solver (Section 3.1), the hierarchical HFSCIM hardware architecture (Section 3.2), and a FeFET-friendly online learning scheme (Section 3.3).

3.1 Software: CT-SAC

Neural ODE formulation. Standard reinforcement learning often approximates environment dynamics using discrete state transitions, such as $h_{t+1} = h_t + f(h_t, s_t)$. However, this rigid, uniform-sampling paradigm struggles to capture the inherently continuous

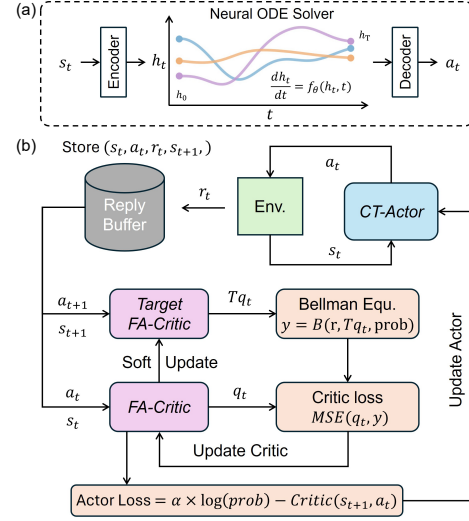


Figure 2: Block diagram of (a) neural ODE solver, (b) continuous-time Soft Actor-Critic.

Algorithm 1: Online Learning Procedure of CT-SAC

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1 Given: temperature  $\alpha$ , discount rate  $\gamma$ , number of batch updates  $B$ 
2 Initialize: CT-Actor  $\theta$ , FA-Critic  $\psi$ , target FA-Critic  $\tilde{\psi} \leftarrow \psi$ , buffer  $D$ , LoRA rank of SRAM  $r$ 
3 for  $t = 0, 1, \dots, T$  do
4    $\triangleright$  Action sampling with hardware noise  $\xi(r)$ 
5    $a_t \leftarrow \pi_\theta(s_t; \xi) \triangleq \text{Dec}_{\theta_d}(\text{ODESolve}(f_{\theta_o}, \text{Enc}_{\theta_e}(s_t), T))$ 
6   Observe  $r_t, s_{t+1}$ , and store  $(s_t, a_t, r_t, s_{t+1})$  in  $D$ 
7   for  $B$  updates do
8     Sample mini-batch  $(s, a, r, s', d) \sim D$ 
9     Sample  $a' \leftarrow \pi_\theta(\cdot | s'; \xi)$ 
10     $\triangleright$  Evaluate bellman target
11     $y \leftarrow r + \gamma(1 - d)[Q_{\tilde{\psi}}(s', a') - \alpha \log \pi(a' | s')]$ 
12     $\psi \leftarrow \psi - \lambda \nabla_{\psi} \text{MSE}(Q_{\psi}(s, a), y)$   $\triangleright$  Update critic
13    Sample  $\tilde{a} \leftarrow \pi_\theta(\cdot | s; \xi)$ 
14     $\theta \leftarrow \theta - \lambda \nabla_{\theta} (\alpha \log \pi(\tilde{a} | s) - Q_{\psi}(s, \tilde{a}))$   $\triangleright$  Update actor
15     $\alpha \leftarrow \alpha - \lambda \nabla_{\alpha} (-\alpha (\log \pi(\tilde{a} | s) + \mathcal{H}))$   $\triangleright$  Update temp.  $\alpha$ 
16     $\tilde{\psi} \leftarrow \tau \psi + (1 - \tau) \tilde{\psi}$   $\triangleright$  Update target critic
17  end
18   $r \leftarrow r + 1$   $\triangleright$  Entropy reduction via LoRA rank
19 end

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dynamics of physical systems governed by differential equations. To address this, a neural ODE formulates state evolution as a continuous initial value problem. As shown in Figure 2(a), rather than relying on discrete Euler approximations typical of residual networks, neural ODEs directly parameterize the continuous-time vector field (Eq. (2)). By leveraging neural ODE solvers, the network accurately models real-world system dynamics in the continuous-time domain, thereby eliminating the truncation errors inherent to fixed-step discretization.

CT-SAC components. We propose the first continuous-time Soft Actor-Critic (CT-SAC) algorithm, which consists of a continuous-time actor (CT-Actor) and a fully analog critic (FA-Critic), as shown in Figure 2(b). The role of the FA-Critic is to evaluate the action-value function directly in the fully analog domain. Concurrently,

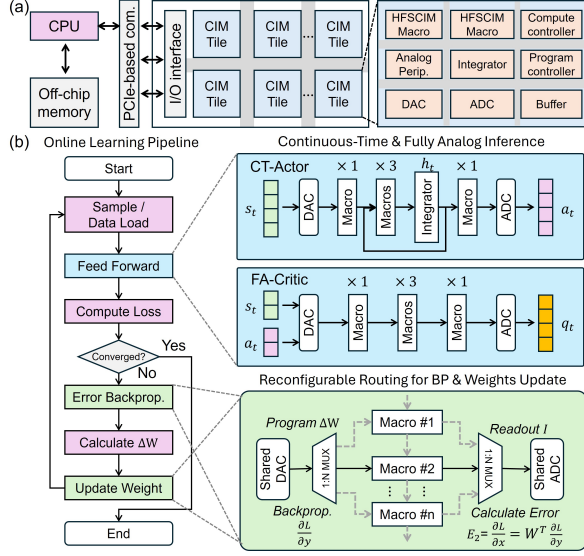


Figure 3: The hierarchy of HFSCIM. (a) Architecture-level design. (b) Online learning pipeline.

the CT-Actor adopts an encoder-decoder architecture integrated with a neural ODE solver. This structure maps discrete state observations s_t into a continuous latent space h_t , evolves this hidden state over time via the parameterized continuous-time vector field, and subsequently decodes it into a continuous action a_t . During environmental interaction, the generated actions, corresponding rewards r_t , and subsequent state transitions s_{t+1} are stored in a replay buffer to facilitate off-policy learning.

Online learning procedure. The learning procedure is outlined in Algorithm 1. Network parameters of CT-SAC are optimized iteratively using mini-batches sampled from the replay buffer $\mathcal{D}$. To mitigate overestimation bias, dual FA-Critic networks (ψ and $\tilde{\psi}$) are updated via temporal-difference learning, minimizing the mean squared error (MSE) against a Bellman target. Crucially, the policy’s stochasticity ξ is inherently derived from the intrinsic noise of the FeFET devices. We dynamically modulate this hardware-induced noise during learning by adjusting the rank of the hybrid FeFET-SRAM CIM macro (detailed in Section 3.3). Initially, a minimal LoRA rank r is used, allowing high device noise to naturally boost SAC entropy and encourage broad state-space exploration. As learning advances, the rank r is incrementally increased to suppress noise, smoothly transitioning the policy from stochastic exploration to deterministic exploitation (detailed in Section 4.3). Concurrently, the CT-Actor is optimized via policy gradients, the temperature α is tuned to maintain a target entropy $\mathcal{H}$, and the target critic networks are updated via a soft moving average τ .

3.2 Hardware: HFSCIM

Overview. To clearly present the hierarchical HFSCIM architecture, we introduce it across three levels: 1) the architecture-level design and online learning pipeline; 2) the macro-level reconfigurable routing and proposed circuit design for FF and BP; and 3) the array-level hybrid charge- and current-domain analog computation.

Architecture-level design. The proposed HFSCIM architecture, designed to implement CT-SAC, is shown in Figure 3(a). A

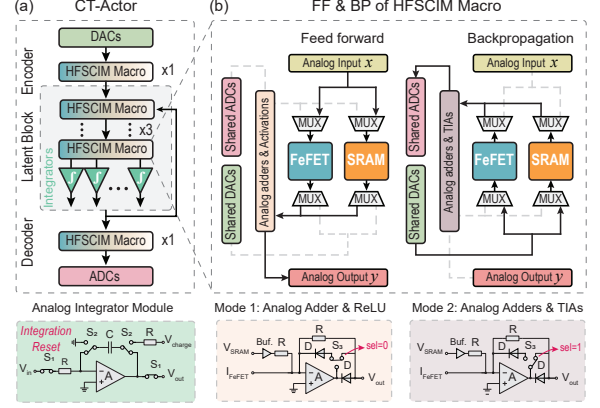


Figure 4: Macro-level design. (a) Continuous-time inference of the CT-Actor. (b) Reconfigurable routing for FF and BP operations.

host CPU communicates via Peripheral Component Interconnect Express (PCIe) with a tiled CIM substrate. Each tile integrates HFSCIM macros for matrix-vector multiplications and dedicated analog integrators for neural ODE dynamics, supported by local mixed-signal analog peripherals. The flowchart (Figure 3(b)) illustrates our pipeline for continuous-time inference and online learning. The CPU handles data sampling and loss/gradient computation, while the CIM substrate offloads the intensive FF, BP, and weight update phases. During FF, the CT-Actor processes states s_t through cascaded macros and integrators to generate actions a_t , while the FA-Critic evaluates state-action pairs to estimate Q-values q_t . This continuous-time analog computation eliminates intermediate analog-to-digital and digital-to-analog (AD/DA) signal conversions, significantly reducing latency and energy cost. During BP, a reconfigurable routing scheme shares digital-to-analog converters (DACs) and analog-to-digital converters (ADCs) across macros via multiplexers to minimize area overhead. This scheme dynamically supports two key operations: (1) efficient analog error backpropagation ($E_x = W^T \frac{\partial L}{\partial y}$), and (2) parallel FeFET weight updates (ΔW) via one-shot programming.

Macro-level design. As shown in Figure 4(a), the CT-Actor executes continuous-time inference using an architecture that consists of an encoder, a solver, and a decoder. By allocating one, three, and one HFSCIM macros to these respective components, the design eliminates power-intensive intermediate DA/AD conversions. Crucially, the numerical integration within the neural ODE solver is implemented by an analog integrator circuit (Figure 4(a), bottom) to evolve the hidden state h_t . This circuit operates in two distinct configurations: (1) **Reset phase** (S_1 open, S_2 closed), where capacitor C is pre-charged to the initial state h_0 ; and (2) **Integration phase** (S_1 closed, S_2 open), where h_t continuously evolves by integrating analog outputs from preceding macros.

A reconfigurable routing design within the HFSCIM macro enables both FF and BP operations (Figure 4(b)). By utilizing multiplexers (MUXes) to dynamically route signals for bidirectional dataflow, the macro operates in two distinct modes: (1) **FF (Mode 1)**, where the arrays perform parallel VMMs. Setting $sel = 0$ opens switch S_3 , configuring the periphery as an analog adder with diode-based ReLU activation. Buffered SRAM output voltages are converted to

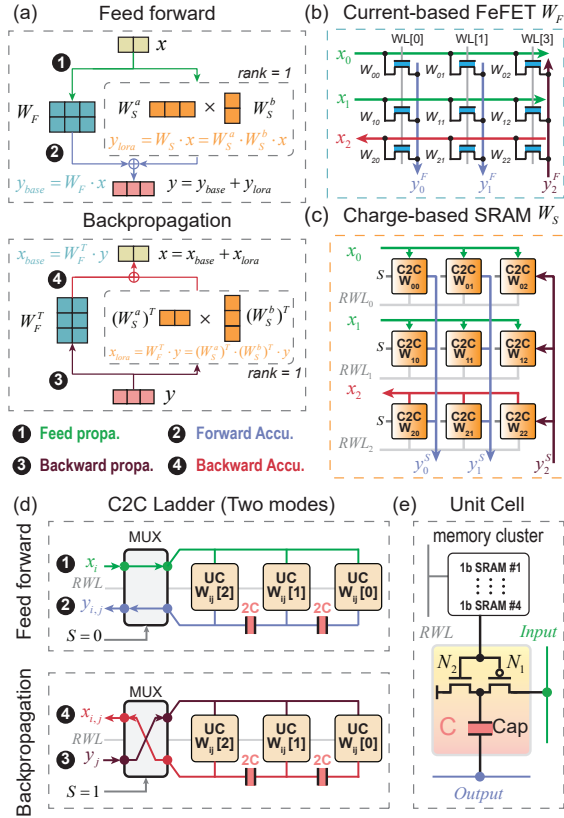


Figure 5: Array-level circuit design: (a) MAC example for FF/BP, (b) current-based FeFET array, (c) charge-based SRAM array, (d) bidirectional C-2C ladder modes, and (e) unit cell.

currents via resistor R and aggregated with FeFET currents at the op-amp summing node (MAC linearity analyzed in Sec. 4.3); and (2) **BP (Mode 2)**, where MUXes reverse the routing for gradient propagation. Setting $sel = 1$ closes S_3 , bypassing the diode to form a linear analog adder and transimpedance amplifier (TIA). Because the ReLU derivative is unity for positive inputs, error gradients pass unchanged and are processed linearly by the TIA, avoiding complex derivative circuitry. This reconfiguration unifies FF and BP operations within the same macro, where the minor MUX and routing overheads are heavily outweighed by the efficiency of peripheral reuse.

Array-level design. Figure 5(a) illustrates the array-level computation of FF and BP within the HFSCIM macro, which partitions the computational workload between long-term weights stored in a current-based FeFET array (W_F) and low-rank compensation weights in a charge-based SRAM array (W_S). During the **FF** phase, ① input broadcasting distributes the vector x to both arrays, followed by ② forward accumulation, which aggregates the partial products to yield the final output: $y = W_F \cdot x + W_S \cdot x = y_{base} + y_{lora}$. Conversely, during the **BP** phase, ③ backward propagation distributes the error signal y through the transposed weight matrices, and ④ backward accumulation computes the input gradients: $x = W_F^T \cdot y + W_S^T \cdot y = x_{base} + x_{lora}$. To physically realize both **FF** and **BP** operations, the FeFET array (Figure 5(b)) employs a compact 1T1F crossbar structure with inherent physical transpose

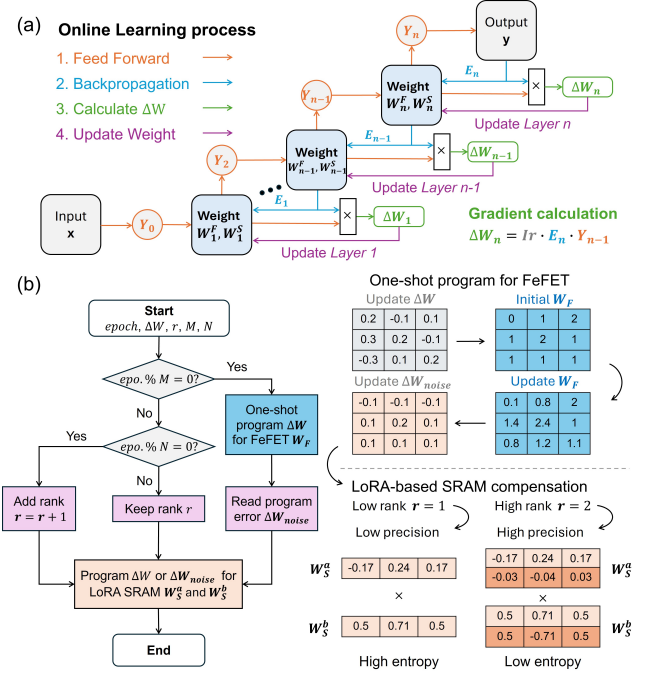


Figure 6: Online learning scheme. (a) Backpropagation dataflow. (b) Macro update flowchart.

symmetry, enabling bidirectional current flow without additional routing overhead.

FF/BP of SRAM Design. To support both FF and BP operations within a unified structure, we propose a charge-based SRAM array. As shown in Figure 5(c), this array employs a 1-to-2 ratioed capacitor ladder (C-2C) CIM scheme to execute multi-bit multiply-accumulate (MAC) operations. Structurally, each C-2C ladder comprises multiple unit cells (UCs), as detailed in Figure 5(e). Each UC integrates a multi-bit SRAM cluster, a pass-gate transistor controlled by a row word line (RWL), and a local capacitor (C).

The operational flow within the SRAM array is governed by a MUX, shown in Figure 5(d), which reconfigures the data path to execute parallel, multi-bit shift-and-add operations directly in the analog domain. During the FF phase ($S=0$), the MUX routes the input forward ①. The C-2C ladder then processes this input to compute the weighted charge summation ② (y_{lora}) according to the following equation:

$$Y_{ij} = V_{MAC_{ij}} = W_{ij} \times X_i = \sum_{k=0}^2 W_{ij}[k] \times X_i = X_i \sum_{k=0}^2 \frac{1}{2^k} \quad (4)$$

Conversely, for the BP phase ($S=1$), the input MUX reconfigures the data path to process the error signal backward ③ through the identical capacitive circuitry. This bidirectional capability enables the efficient computation of the transposed matrix-vector product ④ (x_{lora}), which is mathematically expressed as:

$$X_{ij} = V_{MAC_{ij}} = W_{ij} \times Y_j = \sum_{k=0}^2 W_{ij}[k] \times Y_j = Y_j \sum_{k=0}^2 \frac{1}{2^k} \quad (5)$$

3.3 FeFET-Friendly Online Learning Scheme

Online learning. As illustrated in Figure 6(a), the proposed online learning scheme follows a four-stage iterative process: (1) FF to compute macro activations Y_n , (2) BP to propagate the error signal E_n , (3) gradient calculation $\Delta W_n = I_r \cdot E_n \cdot Y_{n-1}$, and (4) weight updates. To balance the need for non-volatile storage with high-endurance requirements, the weight matrix within each macro employs a hybrid composition of FeFET and SRAM arrays, defined as $W = W_F + W_S$.

Weight update and compensation. Based on our proposed hybrid design, Figure 6(b) details the weight update flowchart tailored to mitigate both the limited endurance and the inherent programming errors of FeFET. First, to preserve device lifetime, the learning scheme primarily updates the SRAM array, governed by a ratio M (set to 20 SRAM updates per FeFET update). When the FeFET update process is triggered, one-shot programming applies the target gradient ΔW to the initial FeFET weights W_F . Second, to compensate for the programming error ΔW_{noise} introduced by device stochasticity, the programming noise ΔW_{noise} is read out and mapped to the SRAM arrays (W_S^a and W_S^b). Crucially, to minimize the hardware overhead of this SRAM-based noise compensation, a LoRA-based method is employed. The compensation fidelity is controlled by the LoRA rank r , which is incremented ($r = r + 1$) every N epochs (set to $N = 20k$). While a lower rank ($r = 1$) yields a high-entropy state with greater policy exploration, dynamically increasing the rank (e.g., to $r = 2$) achieves a low-entropy state, enabling highly precise noise compensation and stable policy convergence.

4 Evaluation and Results

4.1 Experimental Setup

Algorithm benchmarks. We benchmark the proposed CT-SAC algorithm against a suite of established model-free reinforcement learning algorithms [26]: DDPG, PPO, and TD3. To isolate the benefits of the continuous-time formulation, a conventional discrete-time SAC (DT-SAC) is included as a direct baseline. All algorithmic models are implemented in PyTorch 2.1.0.

Hardware platforms. System-level efficiency is assessed by evaluating the HFSCIM architecture across four distinct hardware platforms: a commercial NVIDIA RTX 4090 GPU, a noise-aware analog FeFET-based CIM system, a digital 8-bit integer (INT8) SRAM-based CIM system, and the proposed HFSCIM system. To ensure a fair architectural comparison, all custom hardware baselines and the HFSCIM are modeled at the 28 nm process node.

RL environments. To evaluate the algorithms (Section 4.2) and hardware platforms (Section 4.3), we utilize four standard continuous-control tasks from the MuJoCo benchmark [24]: *Swimmer-v4* (Swv4), *Hopper-v5* (Hopv5), *HalfCheetah-v5* (HCv5), and *Humanoid-v4* (Humv4). Task performance is quantified by the cumulative reward, where a higher value indicates superior policy execution.

Configuration. As summarized in Table 1, the hardware architecture is hierarchically organized from the device to the tile level. The fundamental unit cell (UC) integrates a 2 fF capacitor with SRAM cells to form C-2C ladders. These ladders construct a 32×128 SRAM array featuring a configurable rank from 1 to 32, which operates alongside a 128×128 1T1F array. Together, one FeFET array and two SRAM arrays, combined with requisite peripheral circuitry,

Table 1: HFSCIM hardware summary for CT-SAC.

	Compo.	Num. & Size	Energy (nJ)	Area (μm^2)
UC	Capacitor	1/2 fF	1.62 fJ/act.	0.8
	SRAM cell	4	-	0.048
C-2C	UC/Cap.	4/3	11.34 fJ	5.648
SRAM	C-2C*	$(1-32) \times 128^\wedge$	0.73-23.2	23134.2
FeFET	1T1F	128×128	1.7	302.8
Macro	FeFET/SRAM	1 / 2	3.2-48.1	0.047 mm^2
	Periphery	1	89.3	0.153 mm^2
CT-Actor	Encoder	1	89.6-93.8	0.157 mm^2
	Latent block $^\circ$	3	184.9-274.8	0.399 mm^2
	Decoder	1	2.8-3.2	5141.9
	DAC/ADCs	2/2	13.1/5.9	3046
	others	-	32.4	2216
Tile	CT-Actor	1	328.7-423.1	0.567 mm^2
	FA-Critic	2	309.4-398.4	0.528 mm^2
Chip	Tile	4	2552.4-	4.378 mm^2
			3286.0	

* 50% UC activation; $^\wedge$ Rank = 32; $^\circ$ includes integrator modules.

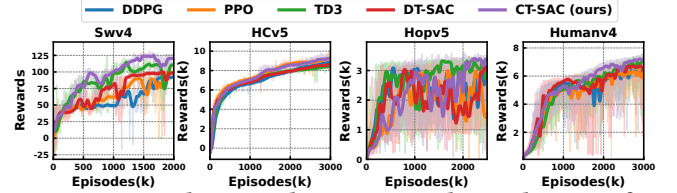


Figure 7: Episodic reward progression during learning for the proposed CT-SAC and baseline methods.

constitute a single computing macro. Building upon this macro structure, the CT-Actor comprises an encoder (one macro), a latent block (three macros incorporating analog integrator modules), and a decoder (one macro). While four DACs and four ADCs are used for external data feeding and readout, this fully analog computing architecture notably eliminates the need for intermediate data converters between internal blocks, significantly reducing peripheral circuitry overhead and power consumption. Finally, the top-level computational tile integrates one such CT-Actor with two FA-Critics.

Methodology. We evaluated the CT-SAC using a comprehensive multi-level simulation framework. Core analog circuits, including the UC, C-2C ladders, hybrid macros, integrators, analog adders, and ReLU activation circuits, were simulated in Cadence Virtuoso. Peripheral components were implemented in SystemVerilog and synthesized using Synopsys Design Compiler. On-chip interconnects and buffers were modeled using CACTI [17]. Finally, we projected system performance and energy metrics from trace-driven simulations, characterizing the 32×128 4-bit integer (INT4) SRAM array using the 28 nm Predictive Technology Model (PTM) and considering realistic noise sources in the FeFET array to accurately assess computational precision.

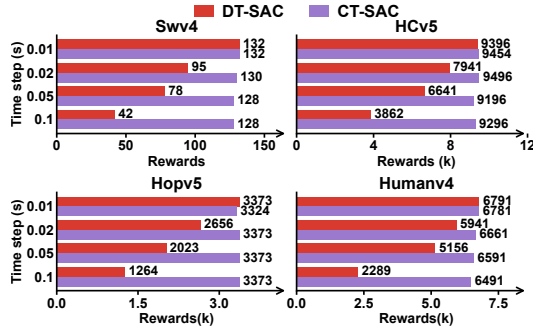


Figure 8: Comparison of temporal robustness across varying inference time steps.

4.2 Algorithm Evaluation

Episodic reward progression. To evaluate the effectiveness of the proposed CT-SAC algorithm, we benchmark it against four representative reinforcement learning baselines across four standard MuJoCo continuous control tasks. All experiments are conducted on an NVIDIA RTX 4090 GPU. To ensure a fair and rigorous comparison of algorithmic capabilities, we maintain strictly consistent configurations across all evaluated policies. Specifically, we use identical network architectures with the same number of trainable parameters, and we standardize the environmental setups, total training steps, and relevant hyperparameters. As shown in Figure 7, CT-SAC demonstrates highly competitive reward convergence across four reinforcement learning environments. For instance, in the Humv4 task, the final cumulative rewards achieved by DDPG, PPO, TD3, DT-SAC, and CT-SAC are 6103, 6252, 6709, 6504, and 7360, respectively. Consequently, the proposed CT-SAC outperforms all evaluated baselines, yielding final reward improvements ranging from 9.70% to 20.60%, including a notable 13.16% enhancement over the standard DT-SAC. This quantitative gain, coupled with the observed stability in the learning trajectories, substantiates the robustness and efficacy of the proposed continuous-time RL framework.

Time-step robustness. To verify that the observed improvement stems from accurate continuous-time modeling, we evaluate the temporal robustness of the trained policies during inference. Both the standard DT-SAC baseline and the proposed CT-SAC are initially trained using a fine-grained time step of $dt = 0.01$ s to approximate real-world continuous-time dynamics. When evaluated at this native resolution, both models exhibit comparable, near-optimal reward values. However, as shown in Figure 8, when the inference time step is relaxed to coarser intervals ($dt \in \{0.02, 0.05, 0.1\}$ s), the cumulative reward of the standard DT-SAC

degrades sharply. This degradation indicates that discrete-time models overfit to the specific temporal resolution observed during training, rendering them highly sensitive to temporal distribution shifts. In contrast, the proposed CT-SAC maintains stable, high-reward trajectories for all evaluated time steps across the four MuJoCo environments. This strong temporal generalization demonstrates that CT-SAC successfully captures the underlying continuous dynamics of the control system, thereby validating its continuous-time formulation.

4.3 Hardware Evaluation

FeFET device-level non-idealities. While FeFETs offer significant advantages for analog computing, including high-density integration and energy-efficient non-volatile storage, their practical application is often constrained by intrinsic device non-idealities, such as write noise, read noise, and limited endurance. The FeFET device itself is not the focus of this work; instead, we use in-house-fabricated HZO-based ferroelectric Schottky-barrier FETs (FE-SBFETs) to validate HFSCIM under realistic device behavior. This device behavior is consistent with prior FeFET reports [13, 30, 36]. To execute analog MAC operations, we utilize the highly linear I-V regime between 0 V and 0.1 V (Figure 9(a)). Operating within this low-voltage region not only minimizes power consumption but also ensures exceptional computational linearity, as evidenced by a coefficient of determination (R^2) of 0.9999. To mitigate endurance degradation by minimizing programming cycles, we employ a one-shot programming scheme, albeit at the cost of introducing more pronounced write noise. Figure 9(b) illustrates the conductance distributions across four distinct states over 100 one-shot programming cycles. These states are programmed by applying varying bit-line (BL) voltages (1.2 V, 1.4 V, 1.6 V, and 2.0 V) under a constant word-line (WL) voltage of 1.4 V, which yields write noise variations ranging from 6.4% to 7.8%. During deployment, the trained weights of the CT-SAC policy, which are bounded within $[-0.6, 0.6]$, are linearly mapped to the conductance values of a one-transistor-one-FeFET array. To preserve device endurance, these weights are subsequently programmed as differential pairs using the aforementioned one-shot programming strategy. Furthermore, to assess the temporal stability of the programmed weights, long-term state retention was monitored across five distinct conductance levels over a duration of 1000 s (Figure 9(c)). These measurements confirm robust non-volatile retention, exhibiting a maximum read noise fluctuation of less than 1.5%.

Macro linearity. To evaluate the robustness of the proposed HFSCIM macro against device-level non-idealities, a linearity variation analysis was conducted using 1,000 Monte Carlo (MC) simulations

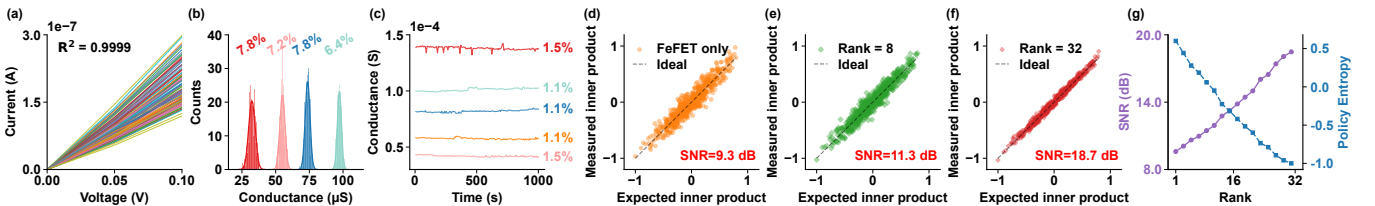


Figure 9: Device non-idealities and HFSCIM performance: (a) I-V sweeps, (b,c) write/read noise, (d-f) MAC accuracy of FeFET only and rank-8/32 HFSCIM, and (g) rank-dependent entropy

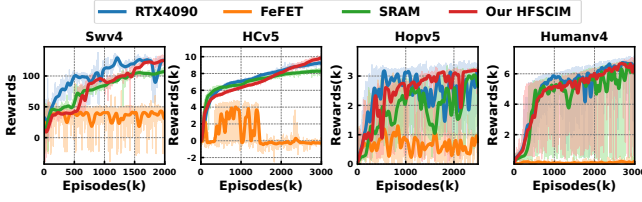


Figure 10: Comparison of training rewards across different hardware platforms.

Table 2: Inference rewards across four hardware platforms.

Task	RTX 4090	SRAM	FeFET	HFSCIM
Swv4	132 ± 4.84	130 ± 6.91	48 ± 20.70	128 ± 7.64
HCV5 (k)	9.32 ± 0.31	9.13 ± 0.72	2.33 ± 0.45	9.23 ± 0.76
Hopv5 (k)	3.37 ± 0.32	3.32 ± 0.46	0.70 ± 0.05	3.31 ± 0.45
Humv4 (k)	7.52 ± 0.79	7.42 ± 1.31	0.20 ± 0.02	7.31 ± 1.18

at the circuit level. The simulation framework models a 128×128 HFSCIM macro. To accurately assess the MAC operation, the input is configured as a fixed 128-dimensional vector with linearly scaled voltages ranging from -0.2 V to 0.2 V. Furthermore, to rigorously capture statistical variations, the MC simulations incorporate specific noise parameters: 8% FeFET write noise and 1.5% FeFET read noise. The output MAC result of the uncompensated, FeFET-only array is evaluated under these conditions, as shown in Figure 9(d), yielding a baseline signal-to-noise ratio (SNR) of 9.3 dB. By integrating the proposed compensation strategy, the computational fidelity improves substantially. Applying compensation ranks of 8 and 32 increases the SNR to 11.3 dB and 18.7 dB, respectively, as shown in Figure 9(e) and (f). This monotonic relationship between the allocated rank and the resulting system SNR is shown in Figure 9(g). Crucially, this hardware-derived SNR directly informs the algorithmic behavior in the CT-SAC framework. Specifically, the policy entropy parameter is defined as proportional to the inverse of the SNR. Consequently, by progressively increasing the compensation rank during the training phase, the framework dynamically modulates the entropy, bridging hardware noise characteristics with the exploration-exploitation trade-off in reinforcement learning.

Online learning rewards under noise. To evaluate online learning robustness, we compare the learning trajectories of the same CT-SAC algorithm. As shown in Figure 10, reward convergence is evaluated across four MuJoCo environments over 10 independent trials, where solid lines and shaded regions denote the mean cumulative rewards and variances, respectively. The FeFET system employs a one-shot programming scheme and relies on analog-to-digital/digital-to-analog (AD/DA) converters for intermediate activation transfers. This configuration inherently introduces substantial write noise on the order of 7% to 8%, alongside significant quantization errors. Consequently, this baseline fails to converge across all evaluated tasks due to the severe accumulation of device-level variations and data conversion inaccuracies. In contrast, the proposed HFSCIM architecture leverages a LoRA precision compensation strategy to dynamically mitigate these analog non-idealities.

Inference rewards under noise. We evaluate the converged policies by averaging cumulative rewards over 50 independent inference episodes across four MuJoCo environments (Table 2). While

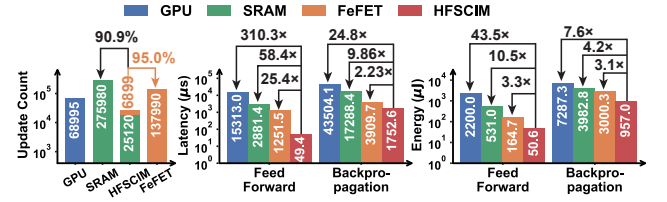


Figure 11: Comparison of average parameter updates, latency, and energy consumption.

uncompensated analog noise in traditional FeFET CIM severely disrupts deterministic policy execution, yielding near-zero rewards, the proposed HFSCIM maintains high computational fidelity. Specifically, HFSCIM achieves average cumulative rewards of 128, 9.12k, 3.31k, and 7.31k for the Swv4, HCV5, Hopv5, and Humv4 tasks, respectively. These results nearly match the high-precision RTX 4090 baseline, confirming that our precision compensation strategy successfully preserves algorithmic efficacy during hardware deployment.

Weight update, latency, and energy consumption. Figure 11 evaluates the system-level efficiency for Swv4 across both feed-forward and backpropagation phases. First, HFSCIM significantly accelerates learning, reducing average parameter updates by 90.9% versus the SRAM CIM baseline, while minimizing FeFET weight update frequency by 95.0% compared to the FeFET CIM baseline. Second, in terms of hardware execution, HFSCIM achieves latency speedups of 310.3×, 58.4×, and 25.4× (24.8×, 9.86×, and 2.23×) during FF (BP) compared to the RTX 4090 GPU, SRAM CIM, and FeFET CIM baselines, respectively. Finally, against these same respective baselines, HFSCIM improves energy efficiency by 43.5×, 10.5×, and 3.3× (7.6×, 4.2×, and 3.1×) during FF (BP). These latency and energy savings stem from the proposed HFSCIM architecture, which leverages fully analog continuous-time inference alongside a FeFET-friendly online learning strategy. Specifically, this strategy uses a LoRA-based SRAM module to efficiently manage primary weight updates and error compensation.

5 Conclusion

This paper presents *HFSCIM*, a novel software-hardware co-design architecture that enables high-fidelity continuous-time inference and energy-efficient online reinforcement learning at the edge. By synergistically integrating a continuous-time policy, an FeFET array with LoRA-based SRAM compensation, and an FeFET-friendly weight update scheme, the proposed architecture achieves robust continuous control with minimal energy overhead. We believe this work demonstrates the potential of unifying inference and online learning within hybrid-memory architectures for scalable and adaptive intelligence in resource-constrained systems.

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